Experiment 12

Nmap to discover live hosts using ARP scan, ICMP scan, and TCP/UDP ping scan in TryHackMe Platform.

Aim

To discover live hosts using ARP, ICMP and TCP/UDP discovery probes on an assigned TryHackMe lab network.

Requirements

TryHackMe Nmap Live Host Discovery room, AttackBox/VPN and an assigned target; Nmap.

Procedure / algorithm

1. Start the TryHackMe lab machine and AttackBox or approved VPN machine. Identify the assigned target/subnet.

2. Preview targets using -sL -n; supply a host, range, CIDR block or -iL list. For 10.10.12.13/29 the block contains addresses .8-.15.

3. Run ARP discovery (-PR -sn) for local Ethernet hosts. Compare ICMP Echo (-PE), Timestamp (-PP) and Mask (-PM) discovery.

4. Run TCP SYN (-PS), TCP ACK (-PA) and UDP (-PU) discovery. Use --disable-arp-ping to compare IP probes on a local link.

5. Observe SYN/ACK or RST for SYN probes; RST for ACK probes; UDP response or ICMP port unreachable for UDP probes. No response is inconclusive.

Complete source / configuration: nmap_commands.sh

```
#!/usr/bin/env bash
set -euo pipefail
TARGETS="${1:?Supply the assigned TryHackMe target or subnet}"
nmap -sL -n "$TARGETS"
sudo nmap -sn -PR "$TARGETS"
sudo nmap -sn -PE --disable-arp-ping "$TARGETS"
sudo nmap -sn -PP --disable-arp-ping "$TARGETS"
sudo nmap -sn -PM --disable-arp-ping "$TARGETS"
sudo nmap -sn -PS80,443 --disable-arp-ping "$TARGETS"
sudo nmap -sn -PA80 --disable-arp-ping "$TARGETS"
sudo nmap -sn -PU40125 --disable-arp-ping "$TARGETS"
# Other target forms: -iL list_of_hosts.txt, an IP range, or CIDR subnet.
# -n skips DNS; -R requests all reverse lookups; --dns-servers selects DNS.
```

Actual output / verification

Not executed in the required lab/tool environment. No actual operational output is claimed.

Expected output - not an execution record

Hosts responding to the permitted discovery probes are listed as up with reasons where available. Results depend on the assigned lab hosts and firewall rules.

Observations

ARP discovery works on local-link IPv4 hosts. Firewalls may suppress probes, so lack of a response is inconclusive. -sn disables port scanning; -Pn skips discovery; DNS resolution can be suppressed using -n.

Result

Complete host-discovery commands and observations are supplied; no TryHackMe/Nmap live scan was executed here.